

MIT 18.06 - Problem Set 7

Section 5.2

- 16) The cofactor of the entry at the $(1,1)$ position is the $\det(F_{n-1})$ and the entry is 1. The cofactor of the entry at $(1,2)$ is the F_{n-2} and the entry is -1 .
So $F_n = 1(F_{n-1}) - 1(-F_{n-2})$
 $= F_{n-1} + F_{n-2}$

32)
$$\begin{aligned} \text{RHS} &= 3F_{2n} - F_{2n-2} \\ &= F_{2n} + F_{2n} + F_{2n-1} + F_{2n-2} - F_{2n-2} \\ &= F_{2n} + F_{2n} + F_{2n-1} \\ &= F_{2n} + F_{2n+1} \\ &= F_{2n+2} \equiv \text{LHS} \end{aligned}$$

$$\therefore F_{2n+2} = 3F_{2n} - F_{2n-2}$$

- 33) Let P_n be the $n \times n$ symmetric pascal matrix. Then:-

$$\begin{aligned} |P_4| &= 20(|P_3|) + 10x_1 + 4x_2 + x_3 = 1 \\ 1 &= 20 + 10x_1 + 4x_2 + x_3 = 1 \\ &\Downarrow \end{aligned}$$

$$0 = 19 + 10x_1 + 4x_2 + x_3$$

~~where th~~ where RHS is the

Section 5.3

$$8) \quad C = \begin{bmatrix} 6 & -3 & 0 \\ 3 & 1 & -1 \\ -6 & 2 & 1 \end{bmatrix}$$

$$ACT:- \begin{bmatrix} 1 & 1 & 4 \\ 1 & 2 & 2 \\ 1 & 2 & 5 \end{bmatrix} \begin{bmatrix} 6 & 3 & -6 \\ -3 & 1 & 2 \\ 0 & -1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$\det(A) = 1c_1 + 1(-c_2) + 4(c_3)$ by cofactors.

Since $c_3 = 1 \times 2 - 2 \times 1 = 0$, therefore changing the value of 4 to 100 or any other value does not change $\det(A)$.

$$28) \quad \begin{matrix} \phi \\ \begin{bmatrix} \sin \phi \cos \theta & \rho \cos \phi \cos \theta & -\rho \sin \phi \sin \theta \\ \sin \phi \sin \theta & \rho \cos \phi \sin \theta & \rho \sin \phi \cos \theta \\ \cos \phi & -\rho \sin \phi & 0 \end{bmatrix} \end{matrix}$$

$$\begin{aligned}
 \det &= \rho \sin \phi \left(\sin \phi \cos \theta \rho \sin \phi \cos \theta + \rho \sin \phi \sin \theta \overset{\sin \theta}{\rho \sin \phi \sin \theta} \right) \\
 &\quad + \cos \phi \left(\rho \cos \phi \cos \theta \rho \sin \phi \cos \theta + \rho \sin \phi \sin \theta \rho \cos \phi \sin \theta \right) \\
 &= \rho \sin \phi \left(\rho \sin^2 \phi \cos^2 \theta + \rho \sin^2 \phi \sin^2 \theta \right) + \\
 &\quad \cos \phi \left(\rho^2 \cos \phi \cos^2 \theta \sin \phi + \rho^2 \sin \phi \sin^2 \theta \cos \phi \right) \\
 &= \rho^2 \sin^3 \phi \left(\cos^2 \theta + \sin^2 \theta \right) + \\
 &\quad \cos \phi \left(\rho^2 \cos \phi \sin \phi \right) \left(\cos^2 \theta + \sin^2 \theta \right) \\
 &= \rho^2 \sin \phi \left(\cos^2 \phi + \sin^2 \phi \right) \\
 &= \rho^2 \sin \phi \equiv J
 \end{aligned}$$

40) Every pair of columns i and j where $i < j$, we take the sum of $(-1)^{i+j+3} = (-1)^{i+j+1}$ multiplied by the det of the first two rows of those columns and the det of the remaining 3×3 matrix.

$$\begin{pmatrix} 2 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & -1 & 2 \end{pmatrix}$$

$$\begin{aligned}
 \det &= \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} \begin{vmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{vmatrix} - \begin{vmatrix} 2 & 0 \\ -1 & -1 \end{vmatrix} \\
 &\quad \times \begin{vmatrix} -1 & -1 & 0 \\ 0 & 2 & -1 \\ 0 & -1 & 2 \end{vmatrix}
 \end{aligned}$$

The rest of the terms are zero so excluded

$$\det = 3 \times 4 - (-2) \times (-3) = 6$$

41) a) The sum of the products of determinants formed from i, j with $i < j$, where i, j are the columns of A for each product, 0.

41) a) The sum over all pairs i, j with $i < j$ of the product of the 2×2 determinant of A and B , where the determinant from A is formed using columns i and j and the determinant from B formed using columns rows i and j .

$$b) A = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 4 & 7 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 1 \\ 2 & 4 \\ 3 & 7 \end{pmatrix}$$

$$\det = \begin{vmatrix} 1 & 2 \\ 1 & 4 \end{vmatrix} \times \begin{vmatrix} 1 & 1 \\ 2 & 4 \end{vmatrix} + \begin{vmatrix} 1 & 3 \\ 1 & 7 \end{vmatrix} \times \begin{vmatrix} 1 & 1 \\ 3 & 7 \end{vmatrix} + \begin{vmatrix} 2 & 3 \\ 4 & 7 \end{vmatrix} \times \begin{vmatrix} 2 & 4 \\ 3 & 7 \end{vmatrix}$$

$$= 2 \times 2 + 4 \times 4 + 2 \times 2$$

$$= 24$$

which matches the determinant of
 $A \times B = \begin{pmatrix} 14 & 30 \\ 30 & 66 \end{pmatrix}$

Section 6.1

19) a) There is a non trivial nullspace because 0 is an eigen value so the rank is not 3, the rank is not 1 because there are multiple non zero eigenvalues, hence rank is 2.

b) ~~$\det(B^*) = 0$ and $\det(B^T B) = \det(B) \times \det(B^T)$~~
 ~~$= 0 \times \det(B^T)$~~
 ~~$= 0$~~

b) B has rank $< n$ so $\det(B) = 0$
 and $\det(B^T B) = \det(B^T) \times \det(B)$
 $= \det(B^T) \times 0$
 $= 0$

c) Not enough info to find

$$\begin{aligned}
 d) \quad Bx &= \lambda x \\
 B^2 x &= \lambda^2 x \\
 (B^2 + I)x &= (\lambda^2 + 1)x \\
 (B^2 + I)^{-1} x &= \frac{1}{\lambda^2 + 1} x
 \end{aligned}$$

So the eigenvalues of $(B^2 + I)^{-1}$ are:-

$$\frac{1}{0^2 + 1} = 1, \quad \frac{1}{1^2 + 1} = \frac{1}{2}, \quad \frac{1}{2^2 + 1} = \frac{1}{5}$$

29) Eigenvalues of A are the diagonal entries i.e. 1, 4 and 6.

Eigenvalues of B are the solutions to:-

$$\det \begin{pmatrix} -\lambda & 0 & 1 \\ 0 & 2-\lambda & 0 \\ 3 & 0 & -\lambda \end{pmatrix} = 0$$

$$-\lambda(2-\lambda)(-\lambda) - 1(2-\lambda)(3) = 0$$

$$(2-\lambda)(\lambda^2 - 3) = 0$$

$$(2-\lambda)(\lambda + \sqrt{3})(\lambda - \sqrt{3}) = 0$$

$$\lambda = 2$$

$$\lambda = -\sqrt{3}$$

$$\lambda = \sqrt{3}$$

The eigenvalues of C are the solutions to:-

$$\det \begin{pmatrix} 2-\lambda & 2 & 2 \\ 2 & 2-\lambda & 2 \\ 2 & 2 & 2-\lambda \end{pmatrix} = 0$$

$$(2-\lambda)((2-\lambda)^2 - 4) - 2(2-\lambda) + 2(2-\lambda) = 0$$
$$-\lambda^3 + 6\lambda^2 = 0$$
$$\lambda^2(-\lambda + 6)$$

$$\lambda = 6 \quad \lambda = 0 \quad \lambda = 0$$

Section 6.2

6) The eigenvalues are the diagonal entries i.e. 4 and 2.

$$\begin{bmatrix} 0 & 0 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \text{when } x = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x'_1 \\ x'_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \text{when } x' = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

So all matrices S that diagonalizes A are the ones with multiples of two columns with a multiple of x in one and a multiple of x' in the other. These same matrices are also the ones that

diagonalize A^{-1} .

16) A_1 is markov so $\lambda_1 = 1$ and
 $\lambda_2 = 0.7 - 1$
 $= -0.3$

$$\begin{bmatrix} -0.4 & 0.9 \\ 0.4 & -0.9 \end{bmatrix} \begin{bmatrix} \chi_1 \\ \chi_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \text{ when } \chi = \begin{bmatrix} 9 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} 0.9 & 0.9 \\ 0.4 & 0.4 \end{bmatrix} \begin{bmatrix} \chi'_1 \\ \chi'_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \text{ when } \chi = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$\text{So } S = \begin{bmatrix} 9 & -1 \\ 4 & 1 \end{bmatrix} \text{ and } \Lambda = \begin{bmatrix} 1 & 0 \\ 0 & -0.3 \end{bmatrix}$$

$$\text{As } k \rightarrow \infty \quad \Lambda^k \rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$S^{-1} S \Lambda^k S^{-1} = \begin{bmatrix} 9 & -1 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -0.3 \end{bmatrix}^k \frac{1}{13} \begin{bmatrix} 1 & 1 \\ -4 & 9 \end{bmatrix}$$

$$\text{As } k \rightarrow \infty, S \Lambda^k S^{-1} \rightarrow \begin{bmatrix} 9 & -1 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \frac{1}{13} \begin{bmatrix} 1 & 1 \\ -4 & 9 \end{bmatrix}$$

$$= \begin{bmatrix} 9 & 0 \\ 4 & 0 \end{bmatrix} \frac{1}{13} \begin{bmatrix} 1 & 1 \\ -4 & 9 \end{bmatrix}$$

$$= \begin{bmatrix} 9/13 & 9/13 \\ 4/13 & 4/13 \end{bmatrix}$$

In the columns we have the steady state vector.

37) Columns of S multiplied by
rows of ΛS^{-1}